

# Aspirin for Primary Prevention in Patients With Elevated Coronary Artery Calcium Score: A Systematic Review of Current Evidences



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The 2019 American College of Cardiology and American Heart Association guidelines regarding low-dose aspirin in the primary prevention of atherosclerotic cardiovascular disease (ASCVD) indicate an increased risk of bleeding without a net benefit. The coronary artery calcium (CAC) score could be used to guide aspirin therapy in high-risk patients without an increased risk of bleeding. With this systematic review, we aimed to analyze studies that have investigated the role of CAC in primary prevention with aspirin. A total of 4 relevant studies were identified and the primary outcomes of interest were bleeding events and major adverse cardiac events. The outcomes of interest were stratified into 3 groups based on CAC scoring: 0, 1 to 99, and  $\geq 100$ . A study concluded from 2,191 patients that with a low bleeding risk, CAC  $\geq 100$ , and ASCVD risk  $\geq 5\%$  aspirin confers a net benefit, whereas patients with a high bleeding risk would experience a net harm, irrespective of ASCVD risk or CAC. All other studies demonstrated net benefit in patients with CAC  $\geq 100$  with a clear benefit. CAC scores correspond to calcified plaque in coronary vessels and are associated with graded increase in adverse cardiovascular events. Our review has found that in the absence of a significant bleeding risk, increased ASCVD risk and CAC score correlate with increased benefit from aspirin. A study demonstrated a decrease in the odds of myocardial infarction from 3 to 0.56 in patients on aspirin. The major drawback of aspirin for primary prevention is the bleeding complication. At present, there is no widely validated tool to predict the bleeding risk with aspirin, which creates difficulties in accurately delineating risk. Barring some discrepancy between studies, evidence shows a net harm for the use of aspirin in low ASCVD risk ( $<5\%$ ), irrespective of CAC score. © 2024 Elsevier Inc. All rights reserved. (Am J Cardiol 2024;220:9–15)

**Keywords:** aspirin, bleeding, coronary artery calcium score, net benefit

The 2019 American College of Cardiology and American Heart Association guidelines have raised questions regarding the role of aspirin in the primary prevention of atherosclerotic cardiovascular disease (ASCVD), citing new data indicating an increased risk of bleeding without a net benefit (class IIb).<sup>1</sup> These guidelines recommend the use of low-dose aspirin in patients with a high ASCVD risk score and low bleeding risk and to consider the coronary artery calcium (CAC) score to identify patients who are likely to benefit by reclassifying their risk and considering primary prevention strategies.<sup>1</sup> CAC scoring, calculated using cardiac computed tomography (CT) scans, quantifies the coronary calcified plaque burden and strongly correlates with major adverse cardiovascular events (MACEs), including cardiovascular mortality, independent of other risk

factors.<sup>2</sup> At present, the CAC score is used to guide statin therapy for reducing cardiovascular risk. However, its applicability in identifying patients who might derive a net benefit from aspirin remains unclear. With this systematic review, we aimed to analyze studies that have investigated the role of CAC in primary prevention with aspirin, aiming to provide clarity to this clinical dilemma.

## Methods

### *Literature search and data extraction*

Our systematic review involved a comprehensive search within the PubMed and Cochrane data sets from inception to June 2023, focusing on studies published in the English language. The search was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Figure 1). Institutional review board approval and written informed consent were not required because all studies included have ethical committee approval and used deidentified data for the analysis.

We aimed to identify studies specifically addressing the use of CAC score as a determinant for aspirin use in the primary prevention of cardiovascular events. To enhance the sensitivity of our search, we combined variants of the words “coronary artery calcium score,” “calcium,” “risk

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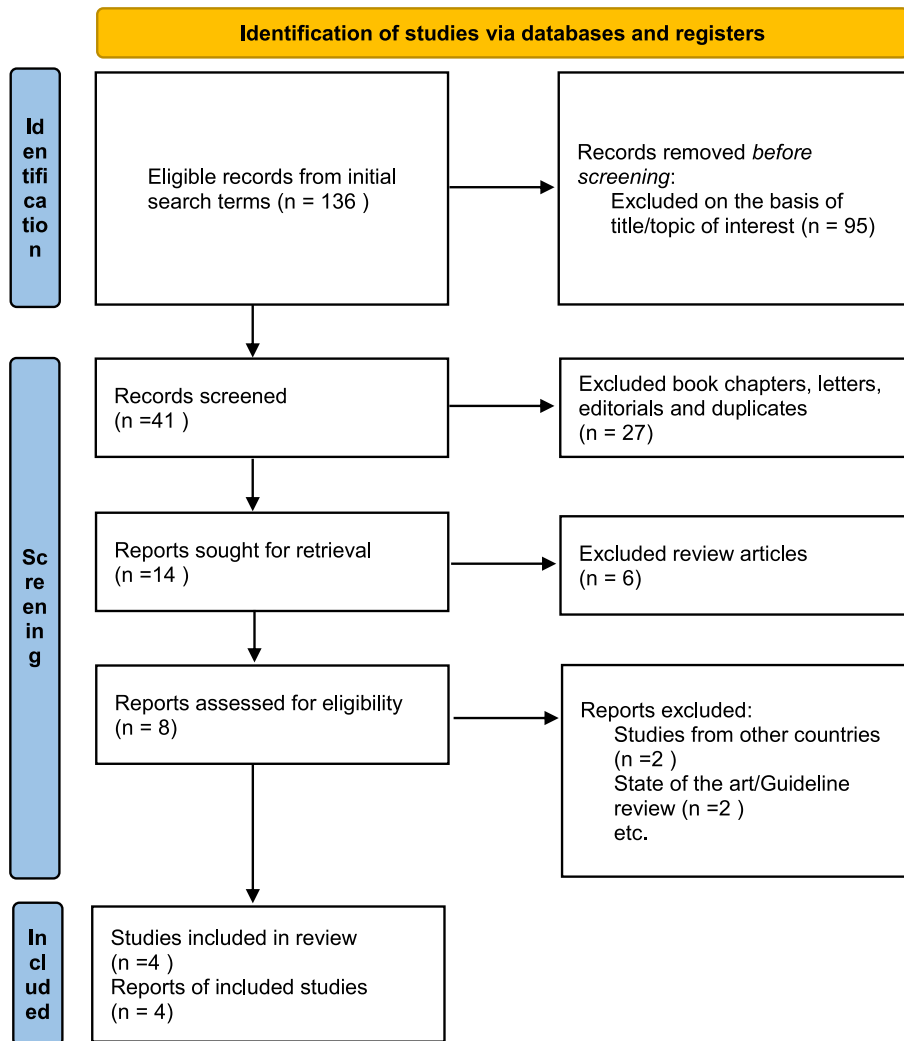


Figure 1. PRISMA flow chart for inclusion and exclusion of the studies for this review.

reduction,” “primary prevention,” “aspirin,” “cardiovascular disease,” “coronary artery disease,” and “risk management” as either keywords or Medical Subject Heading terms. Further references were gathered by scrutinizing the bibliographies of the identified articles. Our predefined eligibility criteria encompassed studies investigating the use of the CAC score as a guiding factor for aspirin use in the primary prevention of cardiovascular events. Notably, we excluded review articles, book chapters, editorials, abstracts, conference presentations, and articles in languages other than English.

After the identification of 4 relevant studies, the data were extracted by 2 independent investigators (AD and HG) and confirmed by a third investigator (RD), all investigators were physicians. Discrepancies were settled by a mutual consensus. The following information was extracted from each article: study name; year of publication; country; number of patients treated; baseline characteristics; and the occurrence of events such as bleeding events, cardiovascular disease (CVD) events, coronary heart disease (CHD) events, and MACE events, as per the definitions within the respective studies. A systematic

review was undertaken because the data could not be subjected to a meta-analysis.

We generated 2 forest plots (Figures 2 and 3) depicting the absolute risk reduction and absolute risk increase in MACE outcomes and bleeding events with aspirin. The study of Limpijankit et al<sup>3</sup> was not included in Figure 2 because of the different method of calculating the effect estimate (mediation analysis as opposed to using effect estimates) and different effect estimate (odds ratio as opposed to absolute risk reduction). Miedema et al<sup>4</sup> was not included in Figure 3 because they used an effect estimate that presumed an identical absolute increase in bleeding risk across all CAC categories. Limpijankit et al<sup>3</sup> was not included in Figure 3 because it did not report on any effect of aspirin on bleeding events stratified by CAC scores. In addition, we did not show the data on occurrence of MACE outcomes and bleeding events when considering the effect of aspirin because it was estimated by calculating the product of baseline MACE incidence and bleeding events and the relative risk estimates used by individual studies. In essence, the effect sizes displayed by the forest plot would have been used to calculate the MACE incidence and bleeding events

## Effect of Aspirin on MACE Outcomes

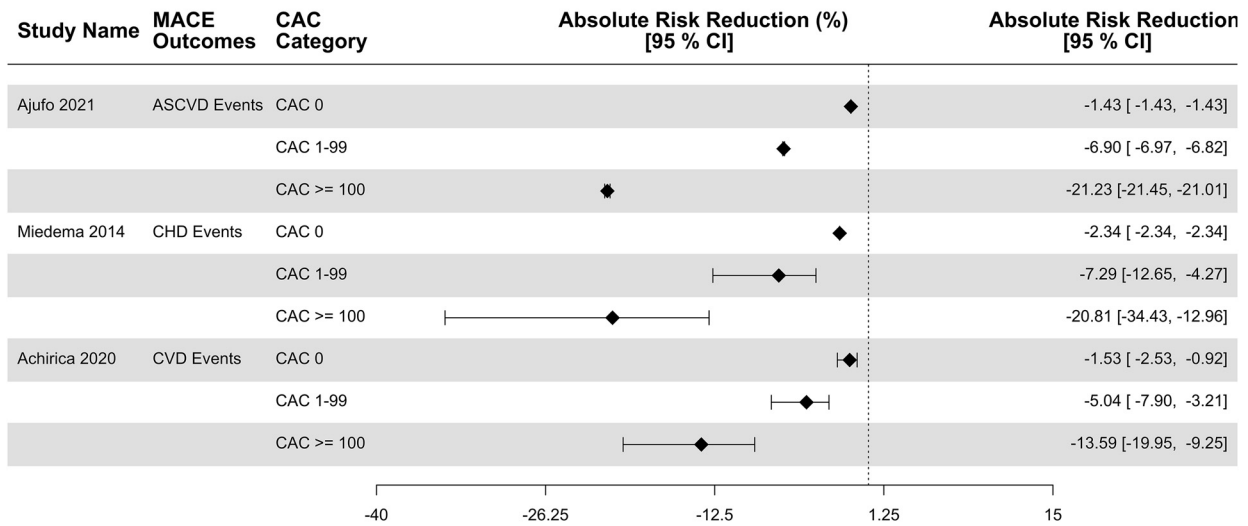


Figure 2. Forest plot depicting absolute risk reduction (%) for MACE outcomes.

after aspirin rather than the other way around, which is usually the case.

## Study outcomes

The outcomes of interest were stratified into 3 groups based on CAC scoring: 0, 1 to 99, and  $\geq 100$ , with a high CAC score defined as  $\geq 100$ . It is worth noting that data from 2 of the included studies further subcategorized the CAC scores into  $\geq 100$  and  $\geq 400$ . However, for the purposes of our systematic review, these 2 strata were combined into a single group representing a CAC score of  $\geq 100$ . The primary outcomes of interests were bleeding events and MACE events. MACE events were described as ASCVD events/CVD events/CHD events/myocardial infarction (MI) events in different studies.

## Results

Total of 4 studies were included for the final review (Table 1). The baseline characteristics of the participants from each study are listed in Table 2. The prospective study

by Ajufo et al<sup>5</sup> included 2,191 participants from the Dallas heart study, examining the association between different CAC score and MACE rates and bleeding rates and exploring the net benefit of aspirin at different CAC scores. The study predominantly represented the population of Dallas County, with an overrepresentation of African-American participants (47%). The prospective studies done by Miedema et al<sup>4</sup> and Cainzos-Achirica et al<sup>6</sup> included participants from Multiethnic Study of Atherosclerosis (MESA) registry. Miedema et al<sup>4</sup> included 4,229 participants to study the use of CAC score in guiding aspirin use for preventing CHD and CVD events. The study excluded all patients with diabetes. Cainzos-Achirica et al<sup>6</sup> included 3,540 participants from the MESA registry, whereas only including patients who were aspirin-naïve aged  $< 70$  years with no high bleeding risk. The study by Limpijankit et al<sup>3</sup> is a single-center, prospective cohort study from Thailand studying 7,849 participants with a predominantly Asian population and studied only MI as a cardiovascular event. Figure 2 depicts a forest plot comparing the absolute reduction in risk (%) for the specific MACE outcomes studied by Ajufo et al,<sup>5</sup> Miedema et al,<sup>4</sup> and Cainzos-Achirica et al<sup>6</sup>

## Effect of Aspirin on Bleeding Events

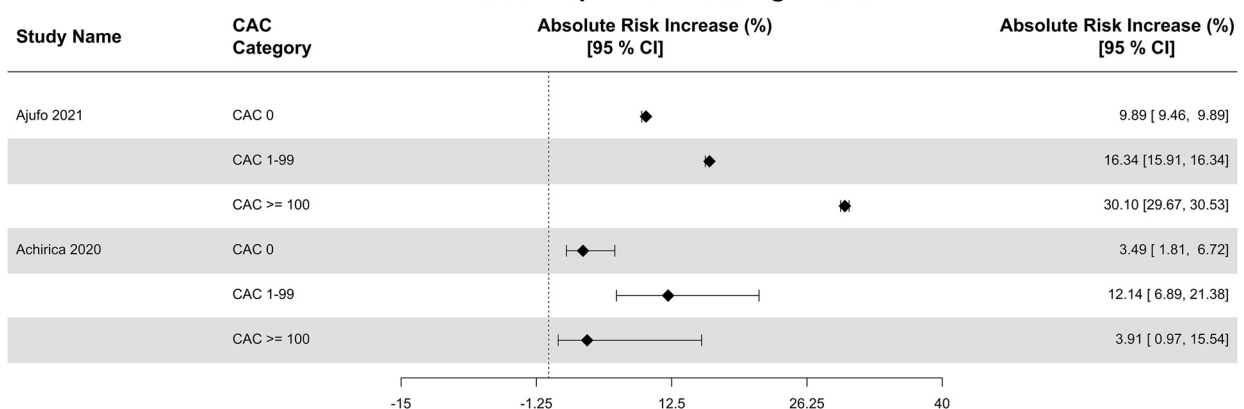


Figure 3. Forest plot depicting absolute risk increase (%) for bleeding events.

Table 1

Name and details of studies included.

|   | Name of the study                                                                                                                                        | Journal and year of publication                                    | Type of study                             | DOI                                                                                                       |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 | Value of Coronary Artery Calcium Scanning in Association with the Net Benefit of Aspirin in Primary Prevention of Atherosclerotic Cardiovascular Disease | JAMA - Oct 2020<br>PMID: 33112372                                  | Prospective population based cohort study | <a href="https://doi.org/10.1001/jamacardio.2020.4939">doi.org/10.1001/jamacardio.2020.4939</a>           |
| 2 | The Use of Coronary Artery Calcium Testing to Guide Aspirin Utilization for Primary Prevention: Estimates from the Multi-Ethnic Study of Atherosclerosis | AHA Circulation CV Quality and Outcomes May 2014<br>PMID: 24803472 | Multiethnic study population              | <a href="https://doi.org/10.1161/CIRCOUTCOMES.113.000690">doi.org/10.1161/CIRCOUTCOMES.113.000690</a>     |
| 3 | Coronary Artery Calcium for Personalized Allocation of Aspirin in Primary Prevention of Cardiovascular Disease in 2019                                   | AHA Circulation - Apr 2020<br>PMID: 32233663                       | Multiethnic study of atherosclerosis      | <a href="https://doi.org/10.1161/CIRCULATIONAHA.119.045010">doi.org/10.1161/CIRCULATIONAHA.119.045010</a> |
| 4 | Causal Relationship of Coronary Artery Calcium on Myocardial Infarction and Preventive Effect of Antiplatelet Therapy                                    | Frontiers in Cardiovascular Medicine Apr 2022<br>PMID: 35571181    | Prospective cohort study                  | <a href="https://doi.org/10.3389/fcvm.2022.871267">doi.org/10.3389/fcvm.2022.871267</a>                   |

across different CAC categories. Figure 3 shows a forest plot depicting the absolute increase (%) in bleeding events with aspirin for Ajufo et al<sup>5</sup> and Cainzos-Achirica et al<sup>6</sup> across different CAC categories. The absolute reduction is risk for Cainzos-Achirica et al<sup>6</sup> was more than the absolute increase in risk of bleeding events at CAC  $\geq 100$ , which shows a net beneficial effect for aspirin (Figures 2 and 3).

#### Clinical outcomes

The outcomes of interest are listed in Table 3. Ajufo et al<sup>5</sup> concluded that in patients at a lower bleeding risk, a CAC score of at least 100 in those at borderline or higher ( $\geq 5\%$ ) 10-year ASCVD risk would experience net benefit from aspirin. In patients at a higher bleeding risk, there

would be net harm from aspirin, irrespective of the CAC and ASCVD risk. Miedema et al<sup>4</sup> calculated the 5-year number needed to treat (NNT5) and the 5-year number needed to harm (NNH5) to identify cardiovascular benefit and bleeding risk, respectively, and stratified results by the 10-year Framingham risk score (FRS). Patients with a CAC score  $\geq 100$  had an estimated net benefit with aspirin, regardless of their traditional risk status (NNT5: 173 for patients  $<10\%$  FRS and 92 for patients  $\geq 10\%$  FRS; NNH5: 442 for a major bleed). Conversely, patients with a 0 CAC score had unfavorable estimations (NNT5: 2,036 for patients  $<10\%$  FRS and 808 for patients  $\geq 10\%$  FRS; NNH5: 442 for a major bleed). Thus, the study concluded that participants with a CAC score of at least 100 had net

Table 2

Baseline characteristics of studies

| Study PMID | Total N | Average Age in Years | Race (%)                                     | Female (%) | BMI        | Smoker (C/F/N)%             | HTN (%) | Diabetes | On Anti-HTN Medication (%) | On lipid lowering therapy (%) | CAC Distribution (%)                       |
|------------|---------|----------------------|----------------------------------------------|------------|------------|-----------------------------|---------|----------|----------------------------|-------------------------------|--------------------------------------------|
| 33112372   | 2191    | 44.4 (9.1)           | W:32.6<br>B: 47.4<br>H: 17.5<br>O: 2.1       | 57         | 29.8 (6.3) | C:26.2<br>F:17.6<br>N: 55.7 | 27.7    | 8.9      | 16.5                       | 4.8                           | 0: 48.5<br>1-99: 44.1<br>$\geq 100$ : 7.4  |
| 24803472   | 4229    | 60.8 (10.2)          | W: 36.7<br>B:27.4<br>H: 22.8<br>Others: 13.1 | 56         | 28 (5.5)   | C:13.6<br>F:34.2<br>N: 52.2 | 38.3    | 0        | 38.3                       | 11.6                          | 0: 55.8<br>1-99: 25.9<br>$\geq 100$ : 18.3 |
| 32233663   | 3540    | 56.5                 | W:34.3<br>B:29.1<br>H: 24.2<br>O: 12.4       | 55.1       | 28.5 (5.7) | C: 15.4<br>N/A              | 10.1    | 10.1     | 27.2                       | 10                            | 0: 62.7<br>1-99: 24.6<br>$\geq 100$ : 12.7 |
| 35571181   | 7849    | 59.3 (8.3)           | Predominantly Asian population               | 64.4       | 24.9 (3.6) | C/F:13.3<br>N: 86.7         | 66.7    | 27.1     | N/A                        | 72.5                          | 0: 48.9<br>1-99: 32<br>$\geq 100$ : 19.1   |

B = Black; BMI = body mass index; CAC = coronary artery calcium; C = current; F = former; H = Hispanic; HTN = hypertension; N = never; O = other; W = White.

Important considerations:

Study 1: predominantly population of Dallas County. Oversampling of African-American population (54%).

Study 2: patients with diabetes excluded.

Study 3: patients with age  $>70$  years and high-risk bleeding features were excluded.

Study 4: single-center prospective study from Thailand, predominantly Asian population.

Table 3  
Outcomes

| PMID     | Outcomes of Interest     | CAC = 0   | CAC = 1-99 | CAC $\geq$ 100 |
|----------|--------------------------|-----------|------------|----------------|
| 33112372 | Total number of patients | 1063      | 967        | 161            |
|          | MACE events (rate)       | 17 (1.3)  | 72 (6.27)  | 34 (19.3)      |
|          | Bleeding events (rate)   | 40 (2.3)  | 59 (3.8)   | 17 (7.0)       |
| 24083472 | Total number of patients | 2361      | 1093       | 775            |
|          | CHD events (rate)        | 31 (1.3)  | 44 (4.05)  | 90 (11.56)     |
|          | CVD events (rate)        | 54 (2.3)  | 73 (6.71)  | 116 (15.02)    |
| 32233663 | Total number of patients | 2219      | 871        | 450            |
|          | CHD events (rate)        | 14 (0.65) | 25 (2.88)  | 45 (9.92)      |
|          | CVD events (rate)        | 31 (1.39) | 40 (4.58)  | 56 (12.35)     |
| 35571181 | Bleeding events (rate)   | 18 (0.83) | 25 (2.89)  | 4 (0.93)       |
|          | Total number of patients | 3842      | 2514       | 1493           |
|          | MI event (rate)          | 28(0.7)   | 45 (1.8)   | 103 (6.9)      |

ASCVD = atherosclerotic; CHD = coronary heart disease; CVD = cardiovascular disease; MACE = major adverse cardiovascular event; MI = myocardial infarction; TIA = transient ischemic attack.

Study 1 (33,112,372): definition of MACE/ASCVD: cardiovascular events associated with hospitalization or death.

Study 2 (24,803,472): CHD events definition: non-fatal MI, resuscitated cardiac arrest and CHD related deaths.

Study 3 (32,233,663): CHD definition: composite of fatal/nonfatal MI and CHD deaths.

CVD definition: composite of fatal/nonfatal MI, fatal/nonfatal stroke, and other CVD death.

CVD events definition = CHD events + nonfatal and fatal stroke, not TIA.

benefit, whereas those with CAC score of 0 had unfavorable risk/benefit profiles with aspirin use for the primary prevention of CVD and CHD. The third study from Cainzos-Achirica et al<sup>6</sup> had a similar outcome showing a net benefit of aspirin with CAC score  $\geq$ 100, especially with score of  $>$ 400 with a lower NNT5 in all ASCVD risk strata. Aspirin was found to be harmful with increased bleeding risk in patients with CAC score 0. Finally, the study from Limpijankit et al<sup>3</sup> demonstrated that in patients with increased CAC scores ( $\geq$ 100, particularly,  $>$ 400), antiplatelet therapy with aspirin significantly reduced the rate of MI compared

with those who did not receive treatment; however, a risk of bleeding was associated with aspirin.

## Discussion

The CAC scores are calculated using cardiac CT scan and indicate the amount of calcified plaque in coronary vessels. They are usually divided into 3 categories with scores CAC 0, CAC 0 to 99, and CAC  $\geq$ 100. Although the categorization of CAC score with these thresholds is arbitrary, these categories have been associated with a graded increase in MACE events.<sup>2</sup> The insights from the CAC Consortium Study indicate that the CAC score is a specific marker of atherosclerosis and can be used beyond calculating mortality risk.<sup>7</sup> Current American College of Cardiology and American Heart Association provide guidelines to identify patients with intermediate ASCVD risk who will benefit from primary prevention with statins.<sup>1</sup> However, guidelines from other organizations, such as the Society of Cardiovascular CT and the National Lipid Association, recommended the use of aspirin for CAC  $>$ 100 in their expert consensus statement.<sup>8,9</sup> However, at present, there are no widely validated tools that can identify patients who will receive a net benefit from aspirin use. With CAC scores, we attempt to identify patients who might have a net gain from aspirin (Figure 4).

The strongest evidence for primary prevention of ASCVD events comes from the study by Limpijankit et al,<sup>3</sup> with a marked decrease in the odds of MI ranging from 3 to 0.56 in patients with CAC  $>$ 400 who took aspirin. However, that study lacks a comparative analysis of bleeding events and only included MI in cardiovascular events. All the studies consistently showed net benefit of using aspirin for primary prevention in patients with CAC score of  $\geq$ 100 for higher ( $>$ 5%) ASCVD risk based on a pooled cohort equation. The benefit increases with an increase in CAC score with maximal benefits in patients with CAC score  $>$ 400. This effect has been consistent throughout all the studies. However, there is a discrepancy of results for the patients with a low risk of 10-year ASCVD ( $<$ 5%). Ajufo

## Summary of findings

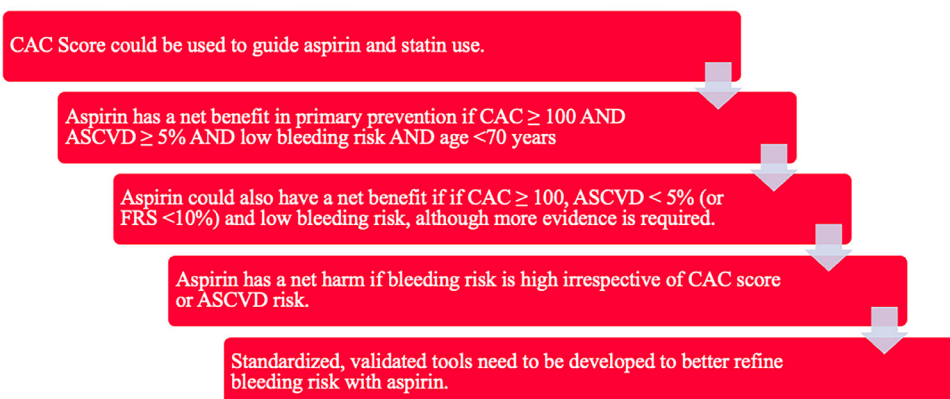


Figure 4. Summary of the findings from this review article.

et al<sup>5</sup> demonstrated that those with a CAC score of  $\geq 100$  but with a low 10-year ASCVD risk ( $<5\%$ ) had a net harm from the use of aspirin. Although the study by Cainzos-Achirica et al<sup>6</sup> showed a net benefit in patients with a CAC score of  $\geq 100$  and ASCVD risk  $<5\%$ , provided they have a low bleeding risk. It may be of interest to note that both the studies used the effect estimates of the same meta-analysis by Zheng et al<sup>10</sup> to gauge the benefits and harms. Therefore, it is peculiar that they have different results in the low-risk ASCVD group, which can be explained by a much wider definition of what constitutes a high bleeding risk in the study by Ajufo et al.<sup>5</sup> In addition, they had increased the baseline bleeding risk with increasing CAC scores, whereas the study by Cainzos-Achirica et al<sup>6</sup> assumed a constant bleeding risk. Other factors that could lead to such differences could be the data collection affected by the study design and differences in age, gender, and racial composition of the study population. Because the study by Ajufo et al<sup>5</sup> relied on hospital data rather than patient reporting (MESA registry), it could potentially represent a more accurate incidence of bleeding and ASCVD events. Moreover, as per the study by Ajufo et al,<sup>5</sup> patients with an ASCVD risk  $>20\%$  have a net benefit with aspirin, even at CAC 0. However, the analysis from the MESA study<sup>4,6</sup> finds no such benefit in patients without diabetes.

The major drawback to the use of aspirin for primary prevention is the bleeding complication.<sup>11</sup> Mainly, hemorrhagic stroke was the most concerning side effect, followed by gastrointestinal bleeding.<sup>12</sup> There is currently 1 aspirin bleeding risk prediction tool; however, it is not widely validated yet.<sup>13,14</sup> The lack of such tools to identify patients at a high risk of bleeding prevents us from accurately delineating groups that could have a net benefit as opposed to net harm. At present, a higher risk of bleeding is identified based on increased age; co-morbidities; and concurrent medication use, such as nonsteroidal anti-inflammatory drugs, history of peptic ulcer disease, alcoholic cirrhosis, simultaneous use of other antiplatelet/anticoagulant agents, bleeding disorders, and so on.<sup>15,16</sup> As per our review, these patients do not have a net benefit, even with a high ASCVD risk and high CAC scores. Thus, with more precise and accurate tools, we can identify groups who have a higher bleeding risk and suggest alternative primary prevention strategies.

The meta-analysis by Zheng et al<sup>10</sup> provides an estimate on the benefit and harm of aspirin use, independent of CAC scores. Thus, if there is any difference associated with CAC score, it would not have been reflected in their results. Although the study from Limpijankit et al<sup>3</sup> demonstrates the benefit of aspirin in the primary prevention of MI in patients with increasing CAC scores, it does not inform us of the harms of using aspirin in those patients. The results of these studies may not be the strongest evidence for guiding use of aspirin in clinical practice; however, they can play an important role in planning of clinical trials that specifically look at the benefits and harms of using aspirin in patients with high CAC scores across different levels of ASCVD risk, especially those with  $>5\%$  risk because that is where the data are consistent across studies.<sup>4–6</sup> This could be a starting point for finding out where we should use aspirin for primary prevention.

## Limitations

The major limitations of our systematic review are the relatively small number of studies and their differences in patient demographics, study designs, and outcomes measured. This can introduce bias and limit the generalizability of our findings. Moreover, 3 of the included studies used effect estimates from meta-analyses to predict the effect of aspirin in their study populations and are predictions rather than actual effects observed in their patient population. Because the estimates generated are from another study's patient population, it is possible that the effects of aspirin in their study groups be different owing to the differences in age, race, and gender. In addition, we could not conduct a meta-analysis of the data because of the differences in the previously mentioned measures because a meta-analysis requires more homogeneity in the data.

## Conclusion

In the systematic review of 4 prospective studies containing the data of 17,809 patients, we identified that a higher CAC score  $\geq 100$  was significantly associated with higher MACE events, and in patients with a low bleeding risk, a CAC score of  $\geq 100$  has a net benefit with the use of low-dose aspirin for primary prevention in patients with high ASCVD risk scores after shared decision-making between patient and a physician. In addition, the use of aspirin for primary prevention proved to be harmful with higher bleeding risks, regardless of the CAC score and ASCVD risk.

## Declaration of competing interest

The authors have no competing interests to declare.

## CRediT authorship contribution statement

**Apoorva Doshi:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization. **Haresh Gandhi:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Kunal N. Patel:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Monil Majmudar:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Rajkumar Doshi:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Conceptualization.

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